Database and Power BI Implementation Record

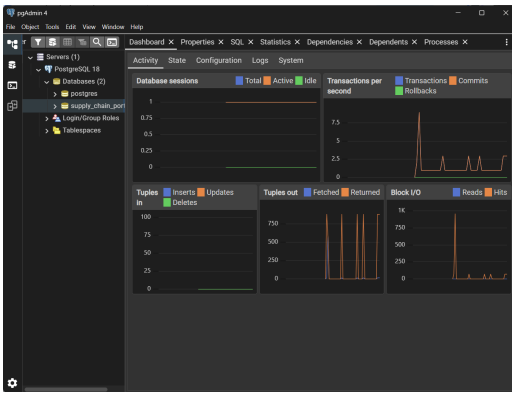
This file records the implementation record and provides a reproducibility reference. Do not commit database passwords or connection strings.

Environment

- PostgreSQL 18
- pgAdmin 4
- Power BI Desktop

Database

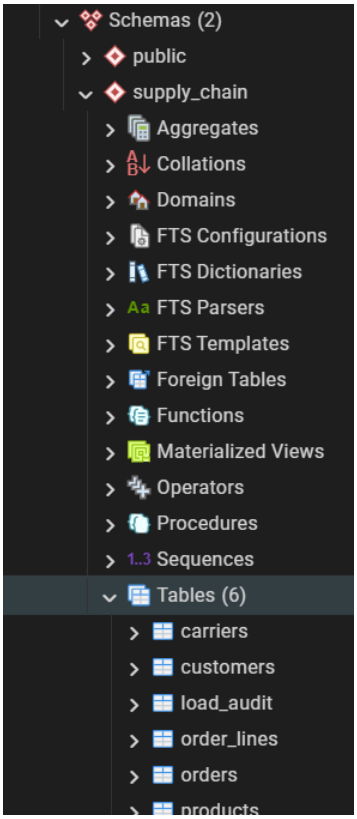
- **Database:** `supply_chain_portfolio`
- Schema: `supply_chain`



Completed Data and SQL Workflow

1. Generated the synthetic tables:
   - `python3 src/generate_synthetic_dataset.py`
2. Ran structural validation:
   - `python3 src/validate_synthetic_dataset.py`
3. Created the database and ran `sql/01_create_tables.sql`.
4. Imported the CSVs manually in foreign-key order:
   - `customers.csv`
   - `products.csv`

    - `carriers.csv`
    - `orders.csv`
    - `order_lines.csv`
5. Executed the remaining SQL files:
    - `sql/02_quality_checks.sql`
    - `sql/03_fulfillment_kpis.sql`
    - `sql/04_segment_diagnostics.sql`
    - `sql/05_dashboard_views.sql`
6. Created the five analytical tables (`customers`, `products`, `carriers`, `orders`, `order_lines`), the `load_audit` support table, and the `order_service` view
7. Manually validated `ORD-000001` and `ORD-000036` for on-time, in-full, OTIF, and cancellation treatment. See ../tests/manual-kpi-validation.md..

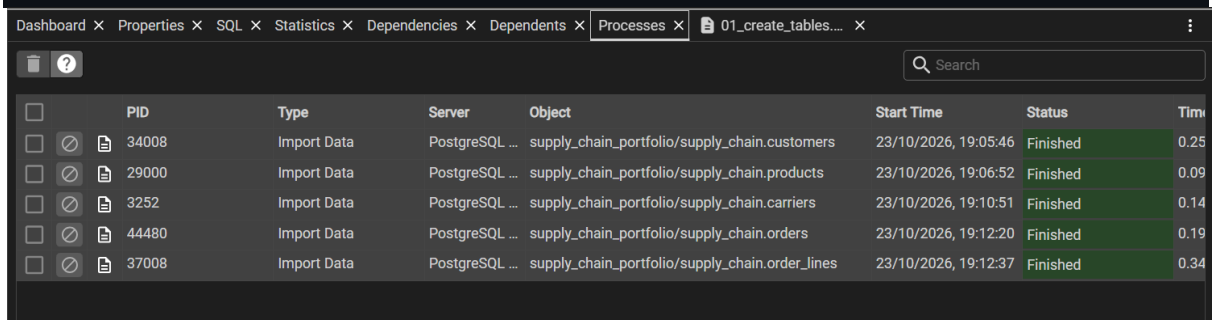


CSV Load Order

1. `customers`
2. `products`
3. `carriers`

4. `orders`
5. `order_lines`

The first three tables are referenced by later order records; order lines depend on both orders and products.



Completed Power BI Work

Connected Power BI to PostgreSQL and imported:

- `supply_chain.order_service`
- `supply_chain.order_lines`
- `supply_chain.customers`
- `supply_chain.products`
- `Supply_chain.carriers`

The relationship model was built on these measures: Submitted Orders, Delivered Orders, On-Time Rate, In-Full Order Rate, OTIF Rate, Cancellation Rate, and Unit Fill Rate.